

Report: Comparison between Mindtooth BP and Mindtooth China

Impedence check

Impedances have been checked by using the two systems. Now, Mindtooth China is able to measure the impedance values, with an offset with respect to the Mindtooth BP amp.

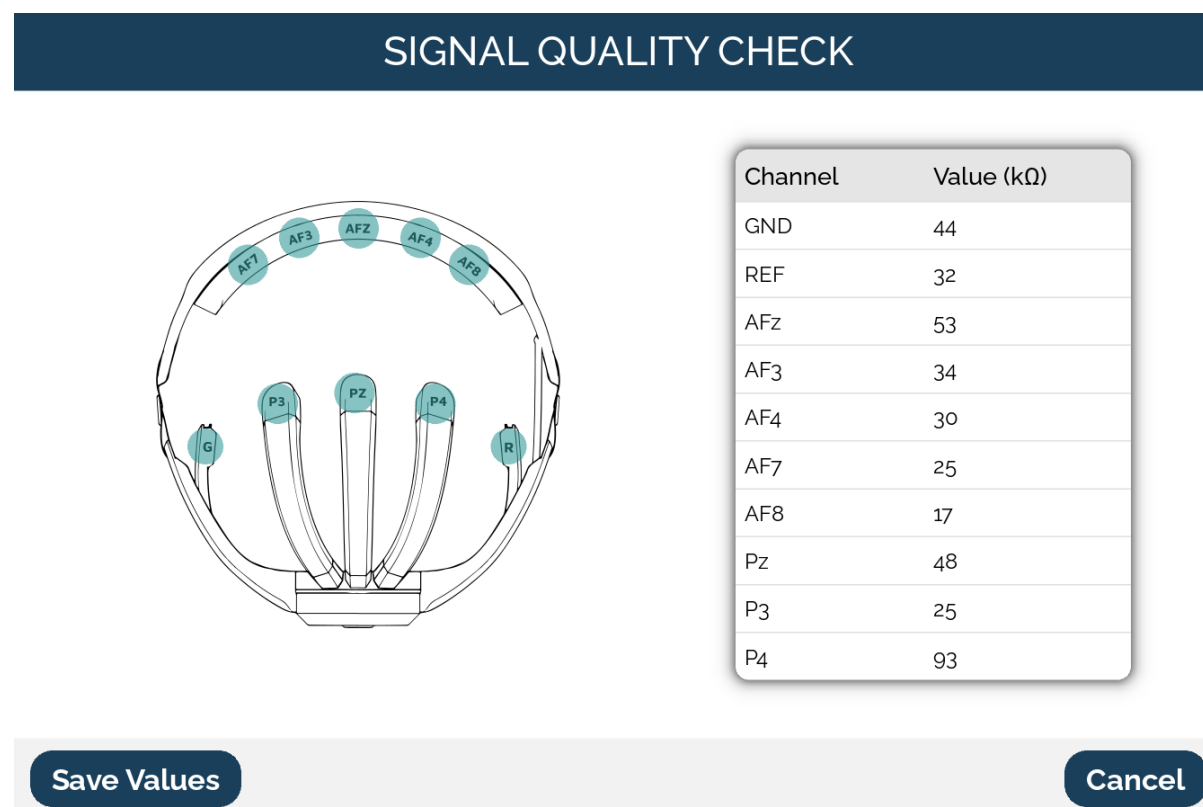


Figure 1: Impedance values of Mindtooth BP

Ground: 44 kohm

Reference: 32 kohm

AFz: 53 kohm

AF3: 34 kohm

AF4: 30 kohm

AF7: 25 kohm

AF8: 17 kohm

Pz: 48 kohm

P3: 25 kohm

P4: 93 kohm

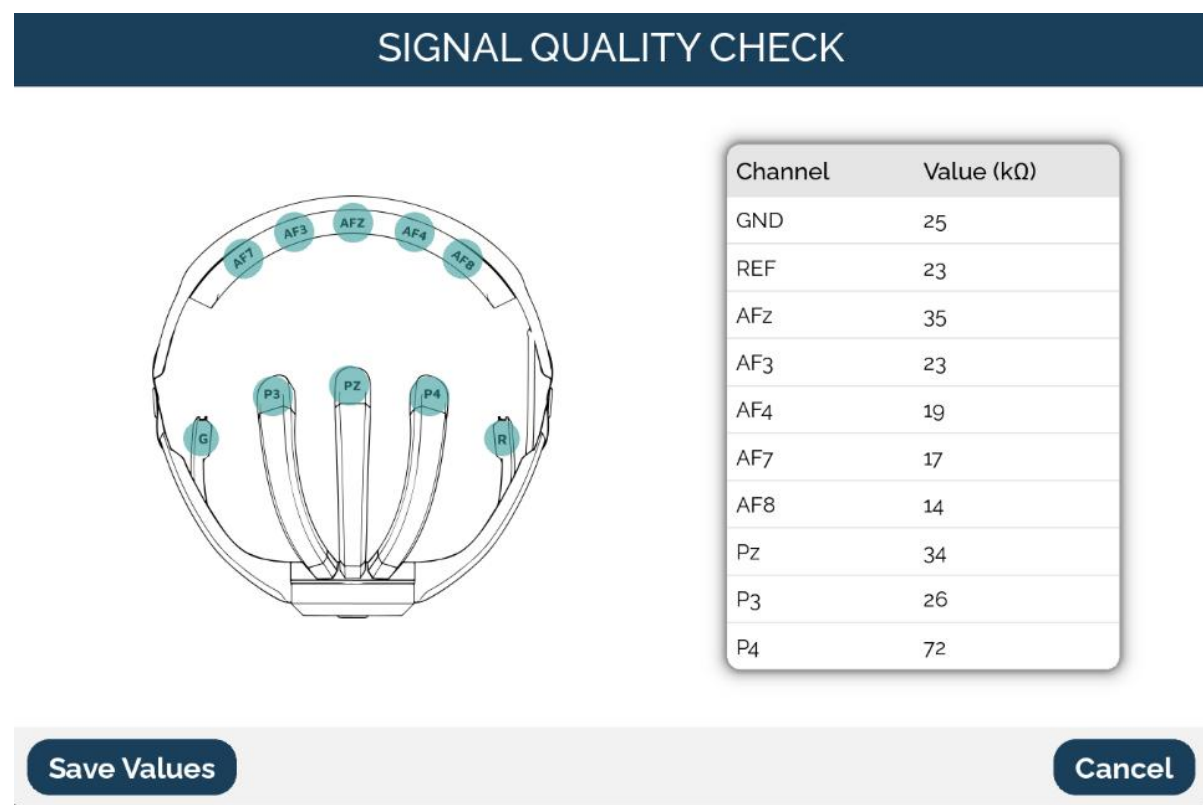


Figure 2: Impedance values of Mindtooth China

Ground: 25 kohm

Reference: 23 kohm

AFz: 35 kohm

AF3: 23 kohm

AF4: 19 kohm

AF7: 17 kohm

AF8: 14 kohm

Pz: 34 kohm

P3: 26 kohm

P4: 72 kohm

It seems that the impedance check of Mindtooth China is underestimated, with an average gain of 1.4. it means that by multiplying impedance values of Mindtooth China amp by 1.4, we should obtain values more similar to Mindtooth BP amp. Of course, this test has been done on 1 person. It has to be repeated several times to confirm this approximation. By the way, the values, even if not fully consistent, are enough to check the signal quality.

Signal similarity

A prototype setup was used in order to test the EEG amplifiers under controlled conditions. In this configuration, all signal electrodes and the reference electrode were electrically short-circuited together, forming a single common input node.

The only electrodes that were kept separated were the ground electrodes, which were connected to different points for the two acquisition systems (two different ear lobes).

This configuration was adopted to ensure that both amplifiers were exposed to the same input signal, thus allowing a direct comparison of their intrinsic behavior (e.g., noise, stability, and baseline characteristics), independently of electrode placement and scalp impedance differences.



Figure 3: Mindtooth headset with double amp

Limitation with dual Mindtooth BP amplifiers

It was not possible to perform a stable simultaneous acquisition using two Mindtooth BP amplifiers. Both devices are equipped with an active **Driven Right Leg (DRL)** circuit. We would like to perform this acquisition with two Mindtooth BP amps, to have a gold standard to be used for the Mindtooth China amp.

The DRL is an active feedback circuit designed to reduce common-mode interference (in particular power-line noise) by sensing the common-mode voltage at the amplifier inputs and reinjecting an inverted signal into the subject through the ground electrode. This improves the overall common-mode rejection of the system but introduces an active control loop between the amplifier and the subject.

When two amplifiers with active DRL circuits are connected simultaneously to the same subject (or to the same electrically common input), the two feedback loops interact and compete with

each other. This interaction leads to instability and low-frequency disturbances in the recorded signals, making reliable acquisition impossible.

Successful acquisition with Mindtooth China amplifier

The acquisition was successfully performed when using one Mindtooth BP amplifier together with the Mindtooth China amplifier. Experimental observations indicate that the Mindtooth China device does not implement an active DRL circuit, or that such a circuit is disabled or passive.

As a result, no competing feedback loops were present, and the two systems could operate simultaneously without introducing low-frequency instabilities or mutual interference.

Limitations due to the absence of a DRL circuit

Although the absence of an active Driven Right Leg (DRL) circuit enabled stable simultaneous acquisition with multiple EEG amplifiers, it may introduce several limitations under less controlled or more challenging recording conditions.

In particular, without a DRL circuit the system may exhibit:

- increased susceptibility to common-mode interference, especially power-line noise (50/60 Hz);
- reduced overall common-mode rejection ratio (CMRR) at the system level;
- higher sensitivity to electrode impedance imbalance and cable movement;
- increased baseline drift in long-term recordings, especially in environments with strong electromagnetic interference.

These effects are typically negligible in compact, battery-powered, wireless headsets with short internal cabling and good electrode contact. However, in environments significant electromagnetic noise, or higher impedances, an active DRL circuit can substantially improve signal robustness and stability.

For these reasons, the inclusion of a configurable or optionally disableable DRL circuit may be beneficial in future implementations, allowing improved noise rejection while preserving compatibility with multi-amplifier or comparative acquisition setups.

Preliminary considerations on differences between the systems

Initial analyses revealed some mismatches between the two EEG acquisition systems. In particular, a gain mismatch was observed between the amplifiers. The Mindtooth China amplifier exhibited signal amplitudes approximately **23.5 times larger** than those recorded by the Mindtooth BP system. This issue was addressed by applying a corresponding gain correction factor and therefore did not represent a critical limitation for the comparative analysis.

A more critical issue was identified in the temporal alignment of the signals. Specifically, the sampling frequency between the two systems was found to be **not fully stable**, leading to a

progressive temporal misalignment over time. As a consequence, the two recordings gradually drift apart, despite being acquired simultaneously. This behavior suggests potential issues at the protocol level, possibly related to the integration between the SDK and the BSR. This aspect is currently under investigation.

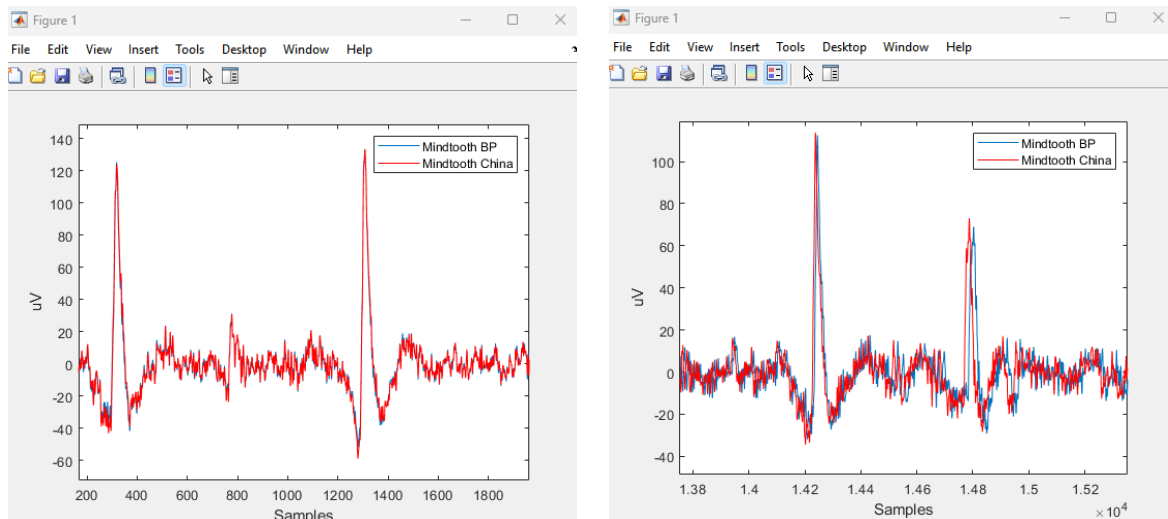


Figure 4: Mismatch between timing of the two recording systems.

Two datasets were acquired: one during **eyes-open** and one during **eyes-closed** conditions. The data were analyzed both in the **time domain** and in the **frequency domain**.

Signal preprocessing

For both systems, the EEG signals were band-pass filtered in the **[1–60] Hz** range prior to analysis.

Time-domain analysis

For the time-domain comparison, the signals were segmented into **1-second non-overlapping windows** for both systems. For each corresponding window pair, the **cross-correlation** between the two signals was computed, and the **maximum correlation value** was retained. This approach was adopted to mitigate the effect of the temporal shift between the two systems.

The results showed **high correlation values across all recorded channels**, indicating strong similarity between the signals despite the temporal misalignment. The correlation coefficients, one for each recording channel (approximated to two decimal places) were:

[0.97, 0.97, 0.98, 0.97, 0.97, 0.95, 0.95, 0.94]

Frequency-domain analysis

For the frequency-domain analysis, a squared-magnitude Fast Fourier Transform (FFT) was computed to obtain the power spectrum. The analysis was performed on 4-second non-overlapping windows. Correlation coefficients between the two systems were then computed over the entire passband and within specific EEG frequency bands, namely Theta (3–7 Hz), Alpha (7–13 Hz), and Beta (13–30 Hz).

The results showed extremely high correlations across all channels. Over the full frequency band, the median correlation values were approximately **1.00 for all channels**, indicating an almost perfect spectral agreement between the two systems.

When analyzing specific frequency bands, similarly high correlations were observed. In the **Theta band**, correlation values were close to unity, with coefficients ranging approximately between **0.99 and 1.00**. In the **Alpha band**, correlations were consistently high, with values around **0.98** across all channels. Slightly lower, yet still strong, correlations were observed in the **Beta band**, with coefficients ranging approximately between **0.93 and 0.96**.

Summary

Despite differences in gain and issues related to sampling frequency stability and temporal alignment, both time-domain and frequency-domain analyses demonstrate a **high degree of similarity** between the EEG signals acquired by the two systems. This suggests that, after appropriate gain correction and alignment strategies, the two amplifiers provide comparable representations of the underlying EEG activity.

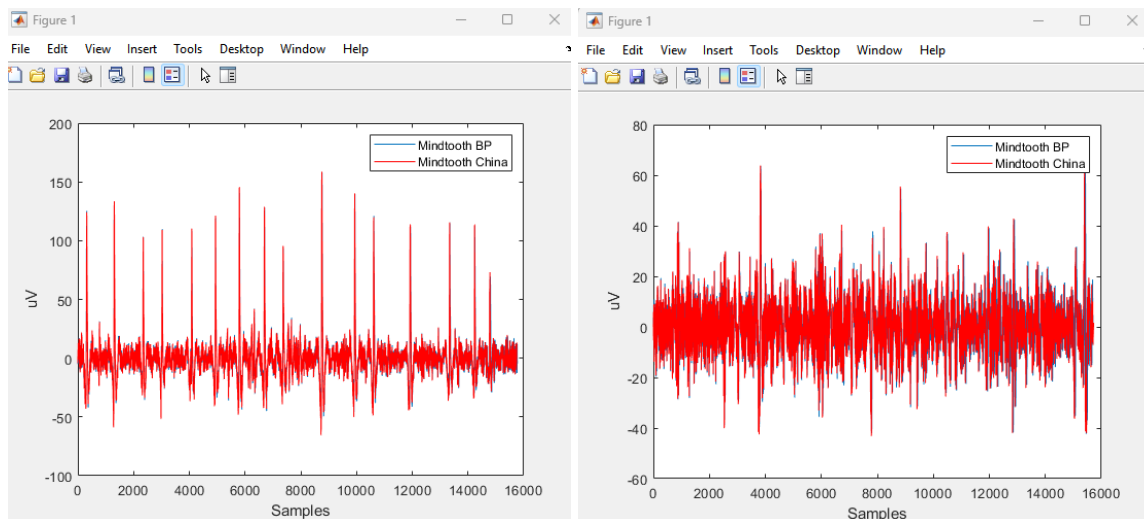


Figure 5: Comparison in time between the two systems, for eyes open (left) and eyes closed (right).

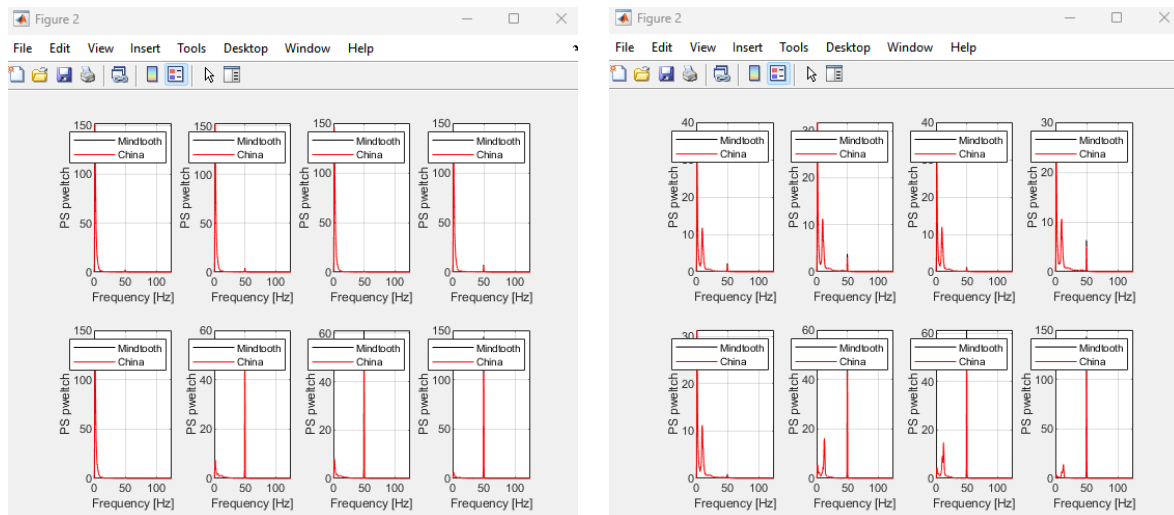


Figure 6: Comparison in frequency between the two systems, for eyes open (left) and eyes closed (right). In particular, in eyes closed can be seen the peak in alpha because of the closed eyes.

Discussion and implementation recommendations

The comparative analysis between the Mindtooth BP and Mindtooth China EEG amplifiers highlighted some differences at both hardware and system-integration levels. These differences can be classified according to their impact on signal quality and system usability.

Gain mismatch

A significant gain difference was observed between the two systems, with the Mindtooth China amplifier producing signal amplitudes approximately **23.5 times larger** than those recorded by the Mindtooth BP device.

This difference does not represent a critical limitation, as it can be effectively compensated through software by applying an appropriate gain correction factor. Once corrected, the two systems show comparable signal dynamics and waveform morphology.

Severity: Low

Recommendation: No immediate action required.

Electrode impedance measurement differences

Differences were observed in the impedance measurements reported by the two systems. However, these values remain primarily **indicative of electrode-skin contact quality** rather than absolute physical quantities. As such, moderate discrepancies between systems do not significantly affect the interpretation of the EEG signal quality.

Severity: Low

Recommendation: No immediate action required.

Temporal misalignment and sampling instability

A critical issue was identified in the temporal alignment between the two systems. Despite simultaneous acquisition, the signals exhibit a **progressive temporal drift**, indicating that the sampling frequencies of the two amplifiers are not perfectly synchronized or stable over time.

This behavior significantly complicates direct signal comparison and may affect applications requiring precise temporal alignment, such as event-related potential (ERP) analysis or multimodal signal fusion. The origin of this issue remains unclear and may be related to the SDK, the underlying DLL implementation, or the communication protocol used between the device and the host system.

Severity: High

Recommendation:

- Investigate sampling clock accuracy and long-term stability.
- Verify timestamping mechanisms at firmware and SDK levels.
- Analyze the data transmission protocol for buffering, packet loss, or resampling effects.
- Consider implementing explicit time synchronization or clock correction mechanisms.

Apparent absence of a DRL circuit

The Mindtooth China amplifier appears to lack an active **Driven Right Leg (DRL)** circuit, or to have it disabled by default. While this did not prevent successful acquisition in the controlled laboratory setup used in this study, the absence of a DRL circuit may significantly degrade signal robustness in less controlled environments.

In particular, without DRL the system may be more susceptible to power-line interference, common-mode noise, and baseline instability, especially in real-world applications involving longer cables, higher electrode impedances, or increased electromagnetic interference.

Severity: Medium to High (application-dependent)

Recommendation:

- Verify whether a DRL circuit is present at the hardware level.
- If present, assess whether it can be enabled or configured via firmware or software.
- If absent, consider implementing a DRL circuit in future hardware revisions, preferably with the option to enable or disable it to ensure compatibility with multi-amplifier or comparative acquisition setups.

Concluding remarks

Overall, while gain and impedance differences between the two systems can be considered minor and manageable, the issues related to temporal misalignment and the apparent absence of a DRL circuit represent more significant limitations. Addressing these aspects would substantially improve the reliability, robustness, and applicability of the Mindtooth China amplifier across a broader range of experimental and real-world EEG acquisition scenarios.

Overall signal quality and final considerations

Despite the differences and limitations discussed above, a key positive outcome of this study is the **high degree of similarity between the signals acquired by the two amplifiers**. Both time-domain and frequency-domain analyses consistently showed **very high correlation values across all channels and frequency bands**, indicating that the two systems capture the same underlying EEG activity with comparable fidelity.

This strong correlation suggests that, from a signal-quality perspective, the Mindtooth China amplifier is capable of providing **high-quality EEG recordings**. Once the identified issues related to temporal alignment and DRL implementation are addressed, the system has the potential to achieve performance levels comparable to the reference Mindtooth BP amplifier, even in more demanding acquisition scenarios.